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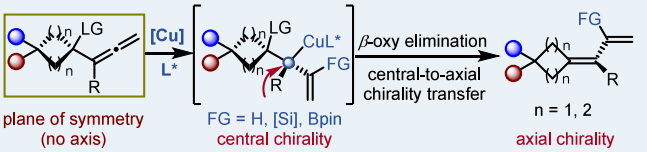
Enantioselective Synthesis of Axially Chiral Alkylidenecycloalkanes via Copper-Catalyzed Functionalization of Acyl Allenols

Baihui Gong,† Qingbin Lu,† Ruikang Sun, Zelin Li, Yihan Chen, Aijun Lin, Hequan Yao,* and Shang Gao*

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ABSTRACT: Axially chiral alkylidenecycloalkanes are an important type of stereoisomer, resulting from the perpendicular arrangement of two pairs of substituents. Despite the considerable progress made in the catalytically enantioselective synthesis of atropisomers, the available methods for accessing axially chiral alkylidenecycloalkanes have been underutilized. In this study, we describe a copper-catalyzed enantioselective synthesis of axially chiral alkylidenecycloalkanes using acyl allenols. The axial chirality is established through a central to axial transfer via an enantioenriched allylcopper species. This reaction features mild conditions, >30:1 regioselectivities, and up to 98% ee. Control experiments reveal that the bulky size of the leaving group has a significant influence on the enantioselectivity of the reaction. The alkene and vinylboronate groups of the products provide versatile platforms for further transformations to enrich the structural diversity of products.



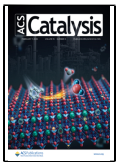
KEYWORDS: *copper catalysis, enantioselective desymmetrization, axially chiral alkylidenecycloalkanes, acyl allenols, allenylic substitution*

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Axially chiral compounds have received significant attention from both synthetic and medicinal communities due to their widespread presence in natural products, chiral ligands, and catalysts.[1] Among these, the catalytical enantioselective synthesis of (hetero)aryl-based C−X (X = C, O, N, B, etc.) atropisomers[2] and allene-based atropisomers[3] has been a hot topic in the synthetic community over the past few decades (Scheme 1a). In comparison, the enantioselective synthesis of axially chiral alkylidenecycloalkanes remains in its infancy due to the challenge arising from the longer distance between the four groups about the chirality axis. At present, the majority of methodologies to access enantioenriched axially chiral alkylidenecycloalkanes focuses on the noncatalytic strategy, i.e., with the aid of stoichiometric chiral reagents.[4] The catalytic methods in this area have been somewhat limited,[5] including enantioselective olefination of 4-substituted cyclohexanones and their analogs,[6] desymmetrization of 4-substituted cyclohexanes and derivatives,[7] [2 + 2] cycloaddition between terminal alkenes and allenes,[8] and deracemization of racemic alkylidenecycloalkanes[9] (Scheme 1b). In view of the remarkable roles of axially chiral alkylidenecycloalkanes in bioactive molecules and chiroptical molecular switches as well as chiral ligands (Scheme 1c),[10] further development of novel methods for synthesizing these architectures is still highly desirable.

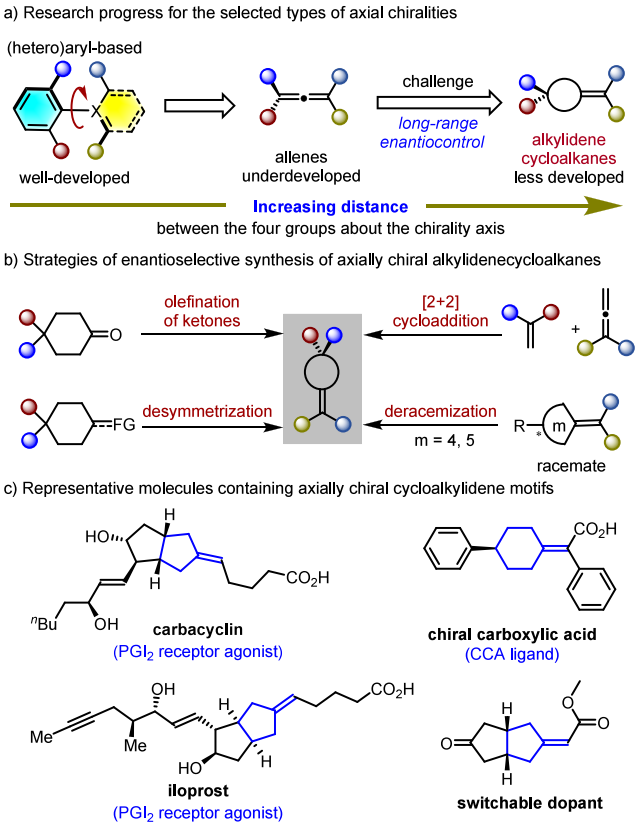
α-Allenols make up a crucial class of allenes that possess a 1,2-propadiene moiety and a hydroxyl group. Due to their π-activated reactivity, α-allenols and their derivatives have found extensive applications in organic synthesis,[11] and various fascinating transformations have been well-established, whereinto, transition-metal catalyzed C1-allylic substitution has emerged as a general strategy to construct allenes bearing a central chirality,[12] or chiral allenes[13] (Scheme 2a, upper panel). When the external nucleophiles attack the C3-position of 2,3-allenol and analogues via transition-metal catalysis, achiral multiply substituted 1,3-dienes would be generated (Scheme 2a, bottom panel).[14] However, to the best of our knowledge, the enantioselective construction of axially chiral alkylidenecycloalkanes using α-allenols and their derivatives remains unknown. We hypothesized that by rational design of allenol derivatives and precise control of the regioselectivity, it would be possible to generate axially chiral alkylidenecycloalkanes by enantioselective C3-substitution.[15] Building upon our research interests in transition-metal-catalyzed enantioselective desymmetrization and atroposelective construction of axially chiral compounds,[16] we present herein a copper-catalyzed enantioselective allenylic substitution of acyl 2,3-allenols (Scheme 2b). With phenylsilane used as the hydride donor, the axially chiral alkylidenecycloalkanes containing a 1,3-dienyl motif are constructed in good yields with excellent enantioselectivities. In addition, copper-catalyzed enantioselective silylation and borylation could also proceed, giving products containing functionalized dienyl motifs.

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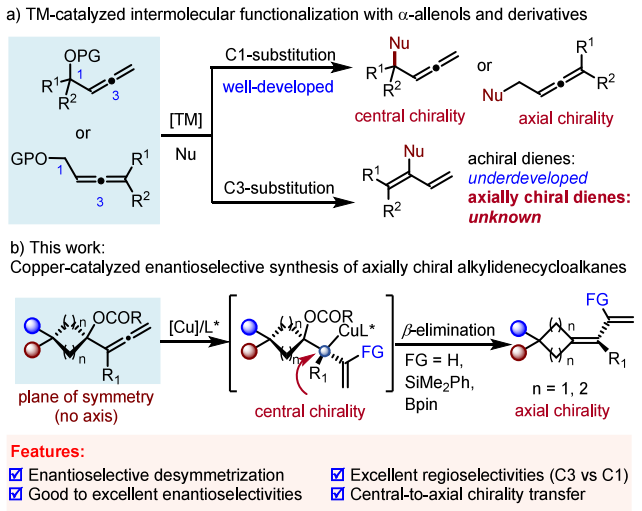


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Scheme 1. Enantioselective Synthesis of Axially Chiral Alkylidenecycloalkanes

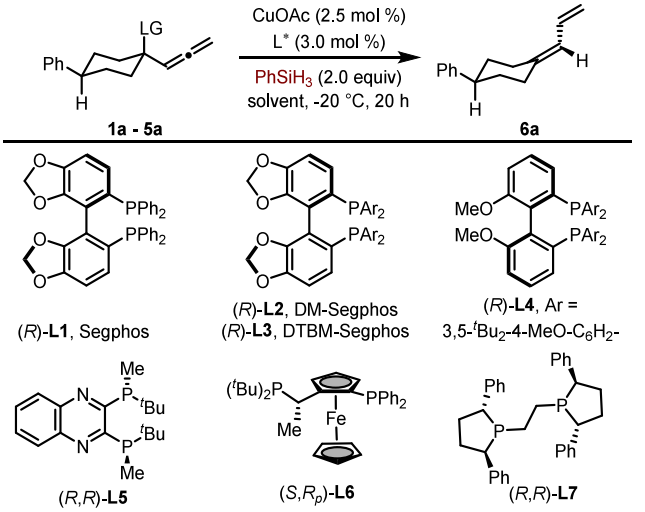


Scheme 2. Transition-Metal Catalyzed Enantioselective Substitution of α-Allenol and Derivatives



We initiated our studies with allenyl acetate **1a** as a starting material. The representative results are summarized in Table 1. In the presence of 2.5 mol % CuOAc and 2.0 equiv of phenylsilane, the reaction proceeded smoothly at −20 °C using various chiral bisphosphine ligands **L1**−**L7**, giving **6a** in 57−86% yields with 11−80% ee (Table 1, entries 1−7). Among these ligands, DTBM-Segphos **L3** exhibited the highest enantioselectivity, with an ee value of 80% (entry 3). To probe the effect of the leaving group on the reaction, we further tested allenol **2a** and its various O-acyl derivatives.

Table 1. Optimization of Reaction Conditions[a-c]



Entry	Ligand	Solvent	LG	Yield (%)	ee (%)
1	**L1**	DME	OAc (**1a**)	61	28
2	**L2**	DME	OAc (**1a**)	71	32
3	**L3**	DME	OAc (**1a**)	69	80
4	**L4**	DME	OAc (**1a**)	76	75
5	**L5**	DME	OAc (**1a**)	57	70
6	**L6**	DME	OAc (**1a**)	86	−11
7	**L7**	DME	OAc (**1a**)	73	51
8	**L3**	DME	OH (**2a**)	29	34
9	**L3**	DME	OCOPh (**3a**)	39	73
10	**L3**	DME	OCO^{i}Pr (**4a**)	80	93
11	**L3**	DME	OCO^{t}Bu (**5a**)	73	96
12	**L3**	DCM	OCO^{t}Bu (**5a**)	29	85
13	**L3**	THF	OCO^{t}Bu (**5a**)	59	96
14	**L3**	Toluene	OCO^{t}Bu (**5a**)	81	97
15[d]	**L3**	Toluene	OCO^{t}Bu (**5a**)	90	98

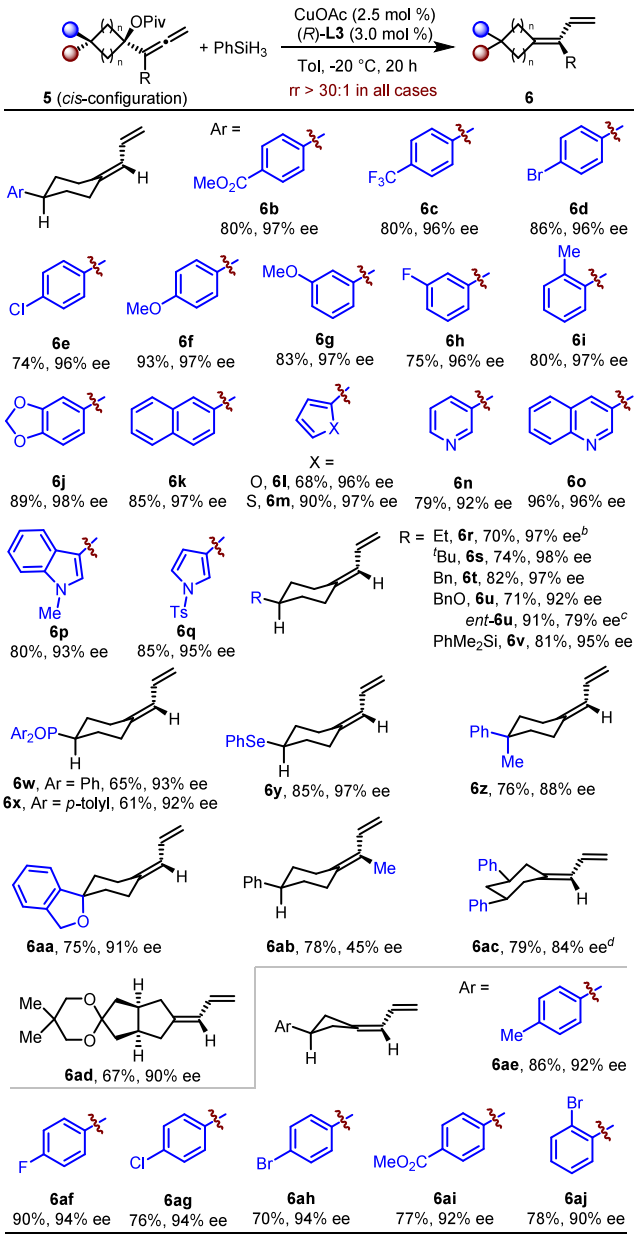
[a]Reaction conditions: **1a**−**5a** (0.2 mmol, 1.0 equiv), CuOAc (2.5 mol %), **L** (3.0 mol %) and $PhSiH_3$ (0.4 mmol, 2.0 equiv) in solvent (0.4 mL) at −20 °C under argon atmosphere for 20 h. [b]Isolated yields were listed. [c]ee values were determined by chiral HPLC analysis on a chiral stationary phase. [d]On 0.5 mmol scale. DME = 1,2-dimethoxyethane; DCM = dichloromethane; THF = tetrahydrofuran.

When allenol **2a** was used, desired product **6a** could only be obtained in 29% yield with 34% ee (entry 8). Switching acetyl allenol **1a** to benzoyl allenol **3a** resulted in a significantly reduced yield, albeit comparable enantioselectivity (entry 9, 39% yield with 73% ee). Introducing a bulkier acyl group led to a marked improvement in the enantioselectivity of **6a**, with ee values of 93−96% and yields of 73−80% (entries 10 and 11). We also observed that the choice of solvent greatly influenced the yield of the reaction (entries 12−14). Among the solvents tested, toluene exhibited the best performance, generating compound **6a** in 81% yield with 97% ee. Furthermore, upon scaling up the reaction to a 0.5 mmol level, there has been a slight enhancement in both the yield and the enantioselectivity of product **6a** (entry 15).

With the optimized conditions established, we next explored the substrate scope of acyl allenols, as presented in Table 2. The pivaloyl allenols derived from 4-arylcyclohexanones, which bear electron-donating or electron-withdrawing groups at the para-, meta-, or ortho-positions of the phenyl ring, all proved suitable for this transformation, delivering **6b**−**i** in 74−93% yields with 96−97% ee. Substrates carrying a piperonyl, 2-

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Table 2. Substrate Scope[a,b,c]



[a]Reaction conditions: **5** (0.5 mmol, 1.0 equiv), CuOAc (2.5 mol %), (*R*)-**L3** (3.0 mol %) and $PhSiH_3$ (1.0 mmol, 2.0 equiv) in toluene (1.0 mL) at −20 °C under argon atmosphere for 20 h. Isolated yields were listed. ee values were determined by chiral HPLC analysis on a chiral-stationary-phase. [b]Using Et_2O as the solvent. [c]With *trans*-**5u** used. [d]Using (*S*,*S*)-BenzP* as the ligand.

naphthyl, or heteroaryl group at the C4-position produced products **6j**−**q** in 68−96% yields with 92−98% ee. Introduction of various alkyl groups led to products **6r**−**t** in 70−82% yields with 97−98% ee. Products **6u**−**y**, containing a series of heteroatom-based group such as BnO-, $PhMe_2Si$-, $Ar_2(O)P$-, or PhSe- group, were successfully synthesized in 61−85% yields with 92−97% ee. Notably, the relative configuration of substrates has a great effect on the enantioselectivity, as well as the absolute configuration of the products. With *trans*-**5u** used as the substrate, *ent*-**6u** was isolated in 91% yield with 79% ee. Furthermore, 4,4-disubstituted cyclohexyl pivalates were also compatible with this transformation, resulting in the formation of product **6z** in 76% yield with 88% ee and **6aa** in 75% yield with 91% ee. Product **6ab** containing an internal tetra-substituted olefin motif could also be obtained in 78% yield, albeit with lower enantioselectivity. Substrates *meso*-**5ac** and *meso*-**5ad** were able to render products **6ac** and **6ad** in 67−79% yields with 84−90% ee. Notably, this transformation allowed for the simultaneous construction of three chiral elements, namely, two C-stereogenic centers and one axial chirality, in a single step.[2a] To enrich the product diversity, a series of cyclobutanone-derived pivaloyl allenols were also examined. Substrates containing phenyl rings with electron-donating or electron-withdrawing groups attached to the cyclobutanol backbones were found to be compatible with the reaction conditions, affording **6ae**−**aj** in 70−90% yields with 90−94% ee. It is worthy of noting that all reactions proceeded with excellent regioselectivities (rr > 30:1) and no allene byproduct was detected.

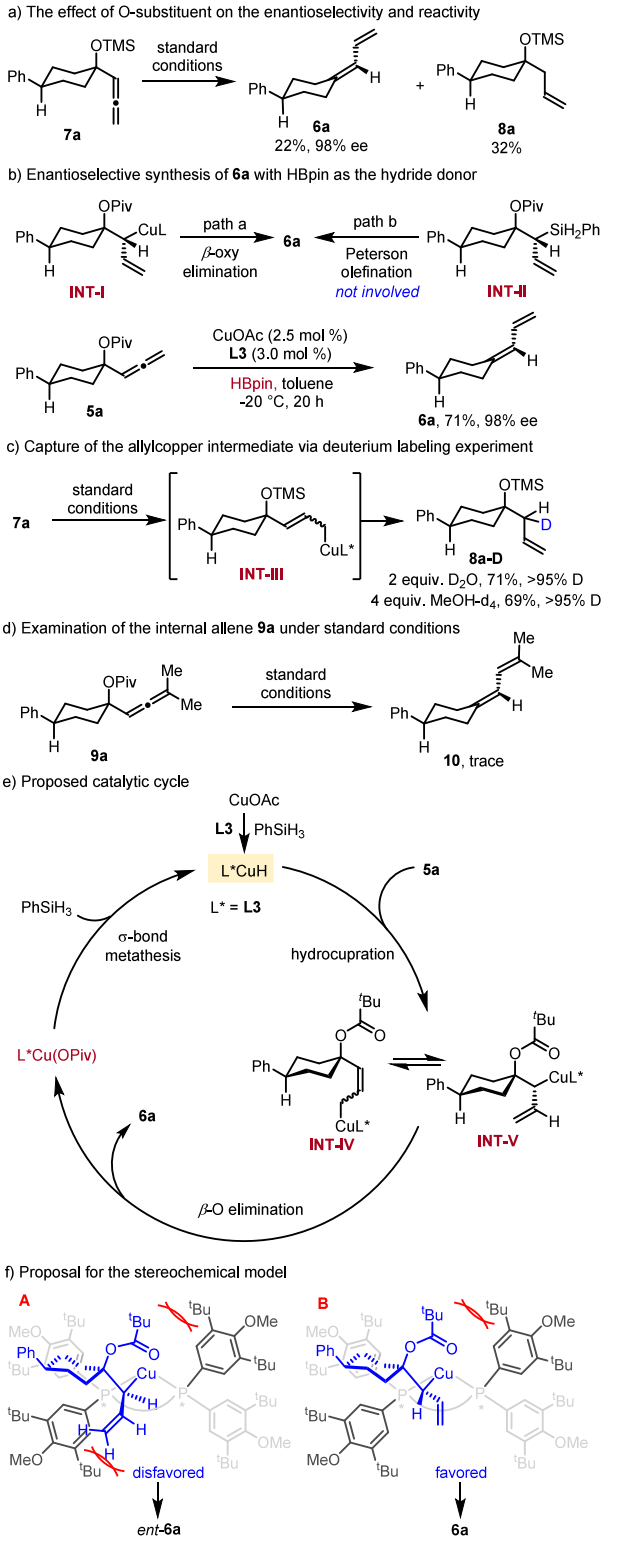
To gain some insights into the reaction mechanism, several control experiments were conducted. As shown in Scheme 3a, with silyl allenol ether **7a** used, the corresponding product **6a** was isolated in 22% yield with 98% ee, accompanied by the formation of partially hydrogenated byproduct **8a** in 32% yield. The excellent enantiopurity of **6a** indicates that the bulky O-substituent is essential to the enantioselectivity, while the acyl group governs the reactivity of substrates to form the chiral axis. At this stage, two possible mechanisms to generate the 1,3-dienyl motif could be proposed including *β*-oxy elimination from allyl copper **INT-I** (path a)[17] and Peterson olefination[18] from **INT-II** (path b) (Scheme 3b, upper panel). Replacing $PhSiH_3$ with HBpin, product **6a** could still be obtained in 71% yield with 98% ee (Scheme 3b, bottom panel), indicating that Peterson olefination (path b) was not involved. In addition, deuterated product **8a**−**D** was isolated in 71% and 69% yields with >95% D in the presence of 2.0 equiv D_2O or 4.0 equiv MeOH-d_4, hinting that allylcopper species **INT-III** was involved (Scheme 3c).[19] Moreover, with pivaloyl allenol **9a** containing two methyl groups used, only trace product **10** was detected (Scheme 3d), which again hints that hydrocupration prefers to occur at the terminal position of allene motif.

Combined with previously reported literature,[20] a possible mechanism is proposed in Scheme 3e. First, CuH species is generated from CuOAc in the presence of **L3** and $PhSiH_3$. Hydrocupration of **5a** provides allylcopper **INT-IV** or **INT-V**. These two allylcopper species could undergo interconversion through a fast 1,3-copper shift.[20,21] *β*-Oxy elimination of **INT-V** will provide **6a** and precatalyst LCu(OPiv). In this step, the axial chirality would be constructed via the central-to-axial chirality transfer. The high enantioselectivity of the substitution of **5a** with $PhSiH_3$ in the presence of (*R*)-DTBM-Segphos (**L3**) was rationalized with two possible transition-state isomers, namely, models **A** and **B** (Scheme 3f). For both models **A** and **B**, a repulsion between the pivaloyl group of **5a** and *t*-Bu group on the aryl unit of (*R*)-DTBM-Segphos (**L3**) is involved. In model **A**, there is a severe repulsion between the vinyl group of the substrate and the sterically bulky *t*-Bu group on the aryl unit of (*R*)-DTBM-Segphos (**L3**). In comparison, the repulsion between the hydrogen atom of substrate **5a** and ligand **L3** is much weaker in model **B**. Finally, *σ*-bond metathesis between LCu(OPiv) and $PhSiH_3$ generates the LCuH species to enter the next catalytic cycle.

At this stage, we were intrigued whether this catalytic system could be further extended to other functionalizations, for

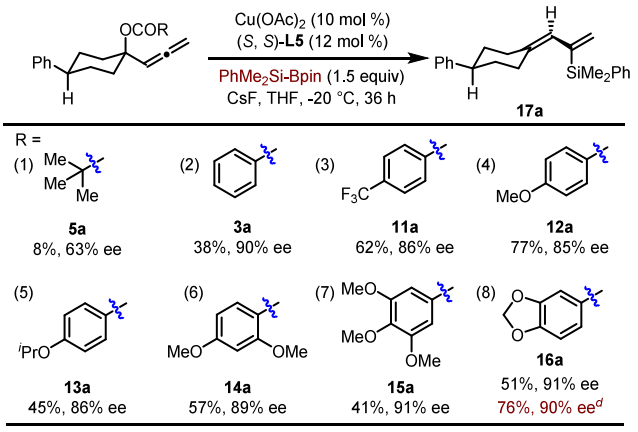
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Scheme 3. Mechanistic Studies



example silylation[22] and borylation.[23] Preliminary investigation showed that (*S*, *S*)-QuinoxP* **L5** is an optimal ligand for this copper-catalyzed silylation of **5a** in the presence of CsF at −20 °C, giving product **17a** in 8% yield with 63% ee (Table 3, entry 1). Since previous results show that the leaving group has a great effect on the enantioselectivity, we further evaluated the leaving groups. Generally, switching the R group from a *tert*-butyl to an aryl group could improve the enantioselectivity

Table 3. Evaluation of Leaving Groups[a,b,c]



[a]Reaction conditions: acyl allenol (0.2 mmol, 1.0 equiv), $PhMe_2SiBpin$ (0.3 mmol, 1.5 equiv), $Cu(OAc)_2$ (10 mol %), (*S*,*S*)-QuinoxP* (**L5**, 12 mol %), and CsF (0.24 mmol, 1.2 equiv) in THF (1.0 mL) at −20 °C under argon atmosphere for 36 h. [b]Isolated yields were listed. [c]ee values were determined by chiral HPLC analysis on a chiral stationary phase. [d]$PhMe_2SiBpin$ (3.0 equiv), 60 h, on 0.5 mmol scale.

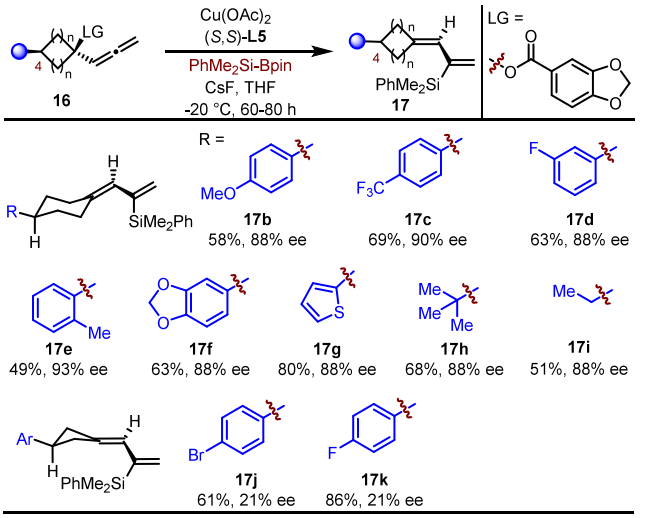
(85−91% ee), while the yields range from 38 to 77% (Table 3, entries 2−8). In particular, with substrate **16a** used, product **17a** was isolated in 51% yield with 91% ee (Table 3, entry 8). By prolonging the reaction time to 60 h, the yield of **17a** could be further improved to 76% yield with a comparable enantiopurity.

With the optimal conditions established, the substrate scope was next examined. As shown in Table 4, a diverse array of acyl allenols containing an aryl group or an alkyl group are suitable substrates for this transformation containing at the C4-positions, giving **17b−i** in 49−80% yields with 88−93% ee. The lower yield for **17e** mainly results from the poor conversion of the substrate. With cyclobutanone-derived substrates used, the corresponding products **17j** and **17k** could only be obtained in 21% ee with 61% and 86% yields, respectively. Furthermore, the copper-catalyzed borylation of acyl allenol **1a** was also tested. With chiral ligand **L3** (DTBM-Segphos) used, the borylated product **18** was isolated in 72% yield with 88% ee (eq 1), which further extended the application scope of this transformation (see Supporting Information for the details of the condition screening). The substrate scope of this borylation was next investigated. For ease of determination of the enantiopurity of the products, the borylated products were further oxidized to the ketones. As shown in Table 5, cyclohexanone-derived allenol esters **1** containing an aryl group or a *tert*-butyl group could participate to this reaction, giving **19b−e** in 62−78% yields with 78−83% ee. Cyclobutanone-derived substrates generated **19f** and **19g** with diminished yields and enantioselectivities.

The practical applicability of this method was demonstrated through a gram-scale reaction and synthetic applications. As shown in Scheme 4a, when the reaction was carried out at 8 mmol, the product **6a** was obtained in 81% yield (1.28 g) with 96% ee. The presence of the 1,3-dienyl motif in the products provided a platform for further transformations[24] (Scheme 4b). Sequential platinum-catalyzed enantioselective diboration/H_2O_2-mediated oxidation of **6a** resulted in the conversion to 1,2-diol **20** in 73% yield with >20:1 dr. Cross-metathesis of **6a** with methyl acrylate using Grubbs second generation

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Table 4. Substrate Scope for Silylation[a,b,c]



[a]Reaction conditions: **16** (0.2 mmol, 1.0 equiv), $PhMe_2SiBpin$ (0.6 mmol, 3.0 equiv), $Cu(OAc)_2$ (10 mol %), (*S*,*S*)-QuinoxP* (L5, 12 mol %), and CsF (0.24 mmol, 1.2 equiv) in THF (1.0 mL) at −20 °C under argon atmosphere for 60−80 h. [b]Isolated yields were listed. [c]ee values were determined by chiral HPLC analysis on a chiral stationary phase.

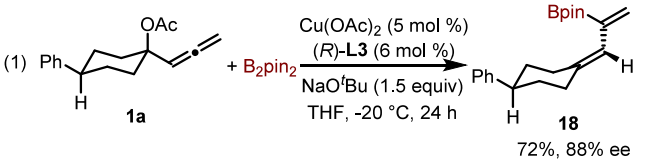
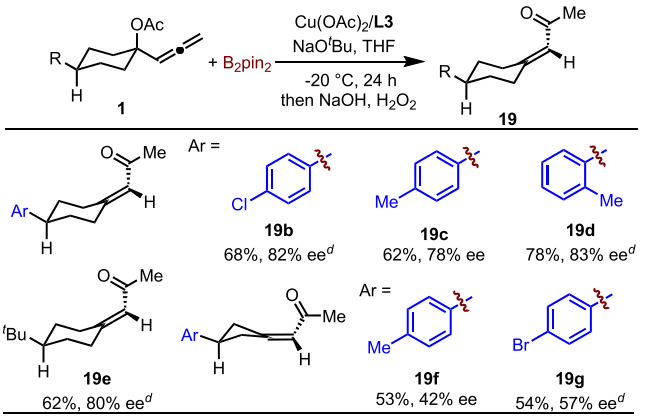


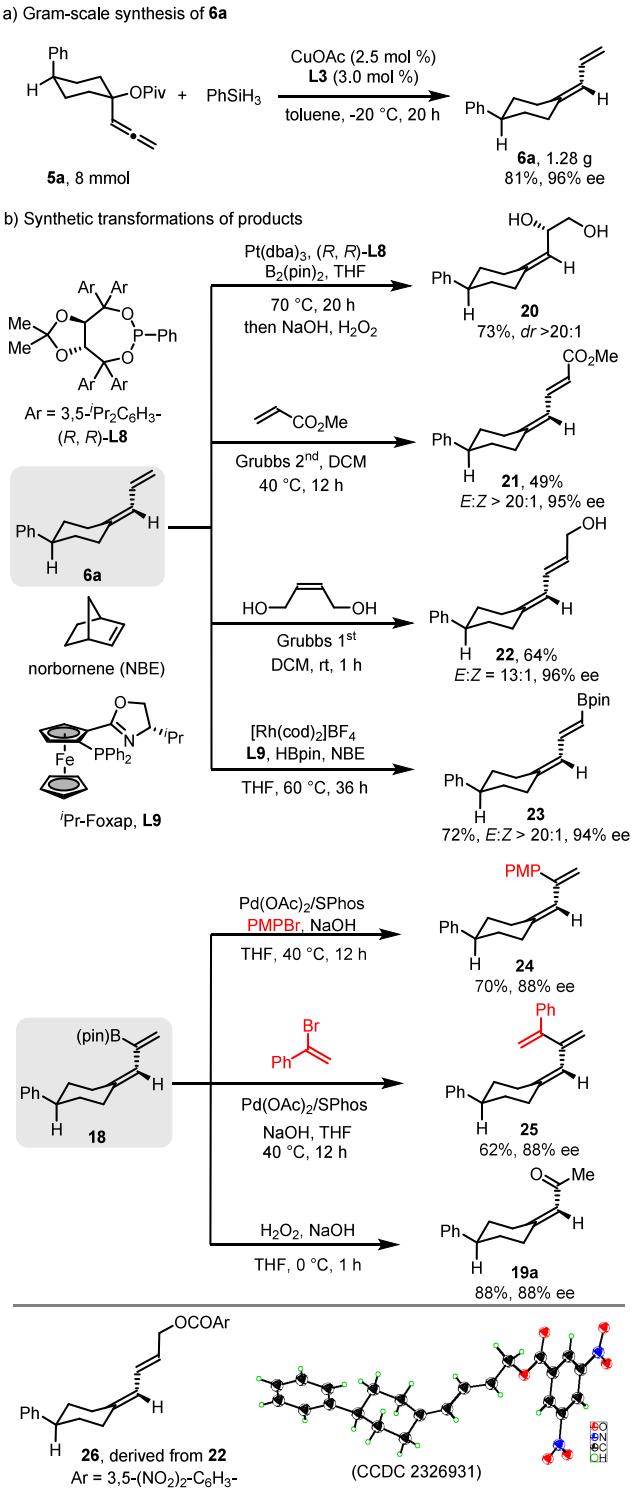
Table 5. Substrate Scope for Borylation[a,b,c]



[a]Reaction conditions: **1** (0.2 mmol, 1.0 equiv), B_2pin_2 (0.4 mmol, 2.0 equiv), $Cu(OAc)_2$ (5 mol %), (*R*)-**L3** (6 mol %), and NaO^{t}Bu (0.3 mmol, 1.5 equiv) in THF (1.0 mL) at −20 °C under argon atmosphere for 24 h; then the reaction mixture was treated with 3 N NaOH and H_2O_2. [b]Isolated yields were listed. [c]ee values were determined by chiral HPLC analysis on a chiral stationary phase. [d]With $Cu(MeCN)_4PF_6$ (5 mol %) as the catalyst in toluene (0.1 M) for 36 h.

catalyst yielded **21** in 49% yield with 95% ee and >20:1 *E*-selectivity. With (*Z*)-2-butene-1,4-diol used as the coupling partner, the corresponding product **22** was isolated in 64% yield with 13:1 *E*-selectivity. Rhodium-catalyzed dehydrogenative coupling reaction between **6a** and HBpin proceeded smoothly to generate vinyl boronate **23** in 72% yield with 94% ee and >20:1 *E*-selectivity. In addition, the vinyl boronate motif of **18** could engage in palladium-catalyzed Suzuki coupling with aryl bromide or alkenyl bromide, delivering **24** in 70% yield and **25** in 62% yield with 88% ee. Oxidation of **18** in the presence of H_2O_2 and NaOH produced **19a** in 88% yield with 88% ee. The absolute configuration of **22** was confirmed through X-ray crystallography analysis using its derivative **26**, and the others were assigned accordingly.

Scheme 4. Gram-Scale Experiment and Further Application of Products



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In summary, we have developed a copper-catalyzed enantioselective synthesis of axially chiral alkylidenecycloalkanes with acyl 2,3-allenols. This reaction features broad substrate scope, mild conditions, and excellent regio- and enantioselectivities. Control experiments show that the choice of acyl group is important to both the yield and enantioselectivity. The terminal alkene and vinyl boronate motifs of the products could participate in a series of downstream transformations. Further application of this catalytic strategy to the enantioselective synthesis of functionalized axially chiral alkylidenecycloalkanes is ongoing in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c06387.

Experimental procedures, characterization, and ^{1}H NMR, ^{13}C NMR, ^{19}F NMR, ^{31}P NMR, and HPLC spectra for all new compounds (PDF)

Data and structure for **26** (CIF)

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Notes

The authors declare no competing financial interest.

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